

GTFS Engineering Synthesis: Architectural Specification and Implementation of the Transit Data Synthesis Engine for Mumbai Metro Line 3

System Architecture and Document Classification Framework

Public transit journey planners function almost exclusively as passive consumers of General Transit Feed Specification (GTFS) datasets¹. This structural dependency creates a critical operational barrier across the Global South, where transit operating agencies rarely publish machine-readable, standards-compliant static or real-time feeds¹. The Transit Data Synthesis Engine (TDSE) fundamentally alters this paradigm by establishing TransitOS as an active producer of standardized transit data¹. Operating on the premise that comprehensive operational and civil engineering literature exists within public planning repositories, the TDSE synthesizes standards-compliant GTFS static archives and estimated GTFS-Realtime streams directly from primary engineering documentation, GIS spatial surveys, and empirical physics modeling¹.

The platform ingests unstructured and semi-structured technical records, transforming them into the Canonical Transit Model (CTM)—a relational, PostGIS-backed schema that provides a unified spatial and operational representation of the transit network¹. To execute this transformation deterministically, the TDSE deploys a document classification protocol that categorizes ingested engineering artifacts across ten discrete categories, labeled Category A through Category I, alongside the synthesized intellectual property tier designated as Category X¹.



Category C: Operations & Headways (DPR Ch. 6, Table 6.7)	
Category D: Rolling Stock Kinematics (DPR Ch. 3, Table 3.4)	
Category E: Signalling & Control (DPR Ch. 3, Table 3.1)	
Category F: Spatial Survey & GIS (DPR Ch. 4, Table 4.1)	
Category G: Fare & Commercial Policy (DPR Ch. 12, Table 12.9)	
Category H: Historical Operations & Dwell Variances	
Category I: Live Observation Probes & Opportunistic Telemetry	
Category X: TransitOS Synthesized Core Assets (CTM, GTFS, GTFS-RT)	

The document classification protocol maps each engineering document directly to its corresponding extraction pipeline and downstream GTFS target entities¹:

- **Category A (Network Topology):** Captures alignment drawings, track chainages, station sequences, curve radii, gradients, and crossover locations¹. In the Mumbai Metro Line 3 (MML3) Detailed Project Report (DPR), Chapter 4 ("Civil Engineering") serves as the Category A baseline, providing the centerline chainage and inter-station distances for the 33.508-kilometer alignment¹.
- **Category B (Station Infrastructure):** Ingests architectural general arrangement drawings, platform layouts, concourse configurations, vertical circulation equipment (elevators, escalators, stairs), and intermodal pathways¹. Chapter 5 ("Station Planning and Intermodal Integration") defines the physical footprints, gate provisions, and pedestrian integration corridors¹.
- **Category C (Operations):** Extracts service spans, peak and off-peak headway policies, first and last train schedules, fleet deployment counts, and turnaround tolerances¹. Chapter 6 ("Train Operation Plan") provides the core operating parameters, establishing the 05:00 to 02:00 service envelope and sectional frequency variations¹.
- **Category D (Rolling Stock):** Quantifies vehicle dimensions, tare and gross vehicular masses, crush carrying capacities, maximum design and operational velocities, and traction acceleration and braking rates¹. Chapter 3 ("System Design - Rolling Stock") governs these kinematic variables¹.
- **Category E (Signalling and Operations Control):** Details train protection systems, Automatic Train Operation (ATO) modes, Computer-Based Interlocking (CBI) arrangements, moving-block headway limits, and design safety margins¹. Chapter 3 ("System Design - Signalling") specifies the Distance-to-Go Automatic Train Protection (ATP) architecture and the 150-second design headway threshold¹.
- **Category F (GIS and Spatial):** Ingests Differential Global Positioning System (DGPS) traverse logs, geodetic control monuments, station platform coordinates, and track vector polylines¹. Section 4.2 ("Engineering Survey") supplies verified DGPS control coordinates for the corridor¹.
- **Category G (Commercial and Passenger Information):** Supplies distance-tiered fare

structures, ticketing media standards, automatic fare collection gate rules, and station commercial directories¹. Chapter 12 ("Financial Analysis") and Chapter 3 ("Automatic Fare Collection") provide the pricing matrices and fare collection rules¹.

- **Category H (Historical Operations):** Captures empirical run times, observed dwell time variances, and section-specific delay parameters derived from operational histories¹.
- **Category I (Live Observations):** Processes real-time observations, such as crowdsourced mobile traces, platform sensor inputs, and opportunistic API logs¹.
- **Category X (TransitOS Synthesized Intellectual Property):** Comprises the derived models generated by the TDSE, including physics-calibrated transit schedules, probabilistic train trajectories, graph-based intermodal transfer topologies, and multi-tier confidence scores¹.

DPR Information Cluster	DPR Chapter Reference	TDSE Category	Canonical Transit Model (CTM) Entity	Target GTFS File
Alignment & Station Chainages	Chapters 0, 4 (Table 0.5, 4.3) ¹	Category A	Station, StationSequence, TrackSegment [cite: 1]	stops.txt, shapes.txt [cite: 1]
DGPS Survey Benchmarks	Chapter 4 (Table 4.1) ¹	Category F	GeodeticPoint, SpatialAlignment [cite: 1]	stops.txt, shapes.txt [cite: 1]
Operating Plans & Headways	Chapter 6 (Table 6.7, 6.8) ¹	Category C	ServicePattern, DispatchRule [cite: 1]	frequencies.txt, trips.txt [cite: 1]
Rolling Stock Kinematics	Chapter 3 (Table 3.4, 3.6) ¹	Category D	VehicleType, KinematicEnvelope [cite: 1]	stop_times.txt (via Physics Engine) ¹
Signalling Safety	Chapter 3 (Table 3.1) ¹	Category E	BlockSection, SignallingMar	frequencies.txt,

Margins			gin [cite: 1]	stop_times.txt [cite: 1]
Station Facilities & Footpaths	Chapter 5 (Table 5.1, 5.2) ¹	Category B	TransferNode , Pathway, StationAccess [cite: 1]	transfers.txt, stops.txt (entrances) ¹
Fare Structures & Distance Slabs	Chapter 12 (Table 12.9, 12.10) ¹	Category G	FareRule, FareZone, TariffMatrix [cite: 1]	fare_attribute s.txt, fare_rules.txt [cite: 1]
Operator Corporate Structure	Chapters 0, 13 (Fig 13.1) ¹	Category G	TransitAgenc y, SystemMetad ata [cite: 1]	agency.txt, feed_info.txt [cite: 1]

Physical Network Topology and Spatial Infrastructure Extraction

Mumbai Metro Line 3 is planned as an entirely underground rapid transit corridor spanning 33.508 kilometers from Cuffe Parade in the extreme south of the Mumbai peninsula to SEEPZ and the Aarey Milk Colony depot in the north¹. The commercial revenue corridor extends 32.546 kilometers between the centerlines of Cuffe Parade and SEEPZ, comprising 27 underground passenger stations¹. A short grade transition north of SEEPZ connects the main running lines to the at-grade maintenance depot at Aarey Milk Colony¹.

The physical alignment was fixed using closed traverse surveying tied to established DGPS monuments¹. Linear measurements were maintained to an accuracy of 4 cm per kilometer, with

leveling closures adhering to a tolerance of $\pm 6\sqrt{K}$ mm, where K represents distance in kilometers¹. Station depth profiles require rail levels to be established at least 15 meters below ground level, descending to 23.5 meters at key underground crossings—such as the Oval Maidan crossing below the proposed Western Railway elevated line and the Mahalaxmi crossing below existing suburban tracks¹. This design maintains a minimum structural rock cushion of 6 meters within competent basalt formations, safeguarding surface structures and shielding the tunnels from utility conflicts¹.

South Terminal: Cuffe Parade (Ch 0 m)	
	(1,000 m)
Badhwar Park (Ch 1,000 m)	
	(600 m)
Vidhan Bhavan (Ch 1,600 m)	
	(685 m)
Churchgate Metro (Ch 2,285 m) [WR Interchange]	
	(817 m)
Hutatma Chowk (Ch 3,102 m)	
	(854 m)
CST Metro (Ch 3,956 m) [CR / Harbour Interchange]	
	(935 m)
Kalbadevi (Ch 4,891 m)	
	(725 m)
Girgaon (Ch 5,616 m)	
	(1,540 m)
Grant Road Metro (Ch 7,156 m)	
	(911 m)
Mumbai Central Metro (Ch 8,067 m) [WR / MSRTC Interchange]	
	(1,149 m)
Mahalakshmi Metro (Ch 9,216 m) [WR / Monorail Interchange]	
	(1,100 m)
Science Museum (Ch 10,316 m)	
	(1,200 m)
Acharya Atrey Chowk (Ch 11,516 m)	
	(1,408 m)
Worli (Ch 12,924 m)	
	(1,555 m)
Siddhi Vinayak (Ch 14,479 m) [MTHL Metro Interchange]	
	(1,277 m)
Dadar Metro (Ch 15,756 m) [CR / WR Interchange]	
	(1,769 m)
Shitla Devi Temple (Ch 17,525 m) [Mahim WR Interchange]	
	(1,781 m)
Dharavi (Ch 19,306 m)	
	(1,965 m)
Bandra Metro (Ch 21,271 m) [Mid-Terminal / Line 2 Interchange]	
	(1,541 m)
Mumbai University, Kalina (Ch 22,812 m)	
	(1,215 m)
Santacruz (Ch 24,027 m)	
	(2,272 m)
CSIA Domestic Airport (Ch 26,299 m) [Terminal 1]	

(1,607 m)
 Sahar Road (Ch 27,906 m)
 (1,052 m)
 CSIA International Airport (Ch 28,958 m) [Terminal 2]
 (871 m)
 Marol Naka (Ch 29,829 m) [Line 1 Interchange]
 (1,396 m)
 MIDC (Ch 31,225 m)
 (1,321 m)
 North Terminal: SEEPZ (Ch 32,546 m)
 (962 m Depot Lead)
 Aarey Depot / Maintenance Facility (Ch 33,508 m)

Geodetic coordinates for all 27 stations were synthesized by merging DGPS engineering survey records from Table 4.1 with spatial geometry from OpenStreetMap¹. Station depth profiles reflect the rail level (RL) and ground level (GL) engineering specifications from the DPR¹.

Sequence	Station Name	DPR Chainage (m)	Inter-Station Distance (m)	Ground Level (m)	Proposed Rail Level (m)	Depth Below Surface (m)	Synthesized WGS 84 Latitude	Synthesized WGS 84 Longitude	Fare Zone ID
1	Cuffe Parade	0	0	+3.43	-12.30	15.73	18.91500	72.82280	Z1
2	Badhwar Park	1,000	1,000	+3.44	-12.00	15.44	18.92200	72.82560	Z1
3	Vidhan Bhavan	1,600	600	+5.14	-16.50	21.64	18.92750	72.82620	Z1
4	Churchgate	2,285	685	+4.00	-19.50	23.50	18.93220	72.82680	Z1

	Metro								
5	Hutat ma Chow k	3,102	817	+6.65	-14.3 5	21.00	18.93 330	72.83 350	Z2
6	CST Metro	3,956	854	+6.68	-10.0 0	16.68	18.94 140	72.83 470	Z2
7	Kalba devi	4,891	935	+5.15	-15.0 0	20.15	18.94 820	72.82 900	Z2
8	Girga on	5,616	725	+5.50	-15.1 0	20.60	18.95 400	72.82 080	Z2
9	Grant Road Metro	7,156	1,540	+2.41	-17.9 0	20.31	18.96 320	72.81 550	Z2
10	Mum bai Centr al Metro	8,067	911	+1.95	-13.2 0	15.15	18.96 960	72.81 940	Z3
11	Maha laksh mi Metro	9,216	1,149	+2.35	-13.0 0	15.35	18.98 270	72.82 420	Z3
12	Scien ce Muse um	10,31 6	1,100	+2.16	-13.1 0	15.26	18.99 020	72.82 130	Z3
13	Acha rya Atrey Chow	11,51 6	1,200	+5.89	-11.0 0	16.89	18.99 720	72.81 810	Z3

	k								
14	Worli	12,924	1,408	+4.52	-11.40	15.92	19.00860	72.81940	Z4
15	Siddhi Vinayak	14,479	1,555	+4.70	-10.70	15.40	19.01680	72.83020	Z4
16	Dadar Metro	15,756	1,277	+4.85	-10.50	15.35	19.02220	72.83860	Z5
17	Shitla Devi Temple	17,525	1,769	+5.71	-9.60	15.31	19.03540	72.84150	Z5
18	Dharavi	19,306	1,781	+4.46	-10.60	15.06	19.04360	72.85500	Z5
19	Bandra Metro	21,271	1,965	+3.56	-11.60	15.16	19.06050	72.86600	Z6
20	Mumbai University, Kalina	22,812	1,541	+3.31	-9.00	12.31	19.07120	72.85960	Z6
21	Santacruz	24,027	1,215	+2.74	-12.30	15.04	19.07920	72.85300	Z6
22	CSIA (Domestic)	26,299	2,272	+5.35	-9.70	15.05	19.09630	72.85420	Z7

23	Sahar Road	27,906	1,607	+13.15	-2.15	15.30	19.10220	72.86530	Z7
24	CSIA (International)	28,958	1,052	+10.37	-5.00	15.37	19.09680	72.87440	Z7
25	Marol Naka	29,829	871	+11.35	-5.00	16.35	19.10980	72.88080	Z7
26	MIDC	31,225	1,396	+25.72	+8.50	17.22	19.12060	72.87530	Z8
27	SEE PZ	32,546	1,321	+29.81	+14.00	15.81	19.13080	72.87970	Z8

To support journey planning, wheelchair routing, and indoor navigation, the TDSE converts each station into parent station nodes (location_type=1), platform child stops (location_type=0), and entrance/exit access portals (location_type=2)¹. Section 5.3 and Tables 5.1–5.2 identify four distinct entrances for Cuffe Parade, five for Vidhan Bhavan, and direct underground connections at Churchgate and CST¹. The synthesized stops.txt file captures this hierarchy:

stop_id	stop_name	stop_lat	stop_lon	zone_id	stop_url	location_type	parent_station	wheelchair_boarding
MM3_01	Cuffe Parade	18.91500	72.82280	Z1	https://mmrcl.com/station/MM3_01	1		1
MM3_01_P1	Cuffe Parade	18.91502	72.82282	Z1		0	MM3_01	1

	Platform 1							
MM3_01_P2	Cuffe Parade Platform 2	18.91498	72.82278	Z1		0	MM3_01	1
MM3_01_E1	Cuffe Parade Entry A1 (WTC)	18.91550	72.82250	Z1		2	MM3_01	1
MM3_02	Badhwar Park	18.92200	72.82560	Z1	https://mmrcl.com/station/MM3_02	1		1
MM3_03	Vidhan Bhavan	18.92750	72.82620	Z1	https://mmrcl.com/station/MM3_03	1		1
MM3_04	Churchgate Metro	18.93220	72.82680	Z1	https://mmrcl.com/station/MM3_04	1		1
MM3_05	Hutatma Chowk	18.93330	72.83350	Z2	https://mmrcl.com/st	1		1

					ation/ MM3_ 05			
MM3_ 06	CST Metro	18.941 40	72.834 70	Z2	https:// mmrcl. com/st ation/ MM3_ 06	1		1
MM3_ 07	Kalba devi	18.948 20	72.829 00	Z2	https:// mmrcl. com/st ation/ MM3_ 07	1		1
MM3_ 08	Girgao n	18.954 00	72.820 80	Z2	https:// mmrcl. com/st ation/ MM3_ 08	1		1
MM3_ 09	Grant Road Metro	18.963 20	72.815 50	Z2	https:// mmrcl. com/st ation/ MM3_ 09	1		1
MM3_ 10	Mumb ai Centra l Metro	18.969 60	72.819 40	Z3	https:// mmrcl. com/st ation/ MM3_ 10	1		1
MM3_ 11	Mahal	18.982	72.824	Z3	https:// mmrcl. com/st ation/ MM3_ 11	1		1

11	akshmi Metro	70	20		mmrcl.com/station/MM3_11			
MM3_12	Science Museum	18.99020	72.82130	Z3	https://mmrcl.com/station/MM3_12	1		1
MM3_13	Acharya Atrey Chowk	18.99720	72.81810	Z3	https://mmrcl.com/station/MM3_13	1		1
MM3_14	Worli	19.00860	72.81940	Z4	https://mmrcl.com/station/MM3_14	1		1
MM3_15	Siddhi Vinayak	19.01680	72.83020	Z4	https://mmrcl.com/station/MM3_15	1		1
MM3_16	Dadar Metro	19.02220	72.83860	Z5	https://mmrcl.com/station/MM3_16	1		1

MM3_17	Shitla Devi Temple	19.03540	72.84150	Z5	https://mmrcl.com/station/MM3_17	1		1
MM3_18	Dhara vi	19.04360	72.85500	Z5	https://mmrcl.com/station/MM3_18	1		1
MM3_19	Bandra Metro	19.06050	72.86600	Z6	https://mmrcl.com/station/MM3_19	1		1
MM3_19_P1	Bandra Metro Platform 1 (Dn Throu gh)	19.06052	72.86602	Z6		0	MM3_19	1
MM3_19_P2	Bandra Metro Platform 2 (Up Throu gh)	19.06048	72.86598	Z6		0	MM3_19	1
MM3_	Bandra	19.060	72.865	Z6		0	MM3_	1

19_P3	Metro Platform 3 (Mid-Term Reversal)	45	95				19	
MM3_20	Mumbai University, Kalina	19.07120	72.85960	Z6	https://mmrcl.com/station/MM3_20	1		1
MM3_21	Santa cruz	19.07920	72.85300	Z6	https://mmrcl.com/station/MM3_21	1		1
MM3_22	CSIA (Domestic)	19.09630	72.85420	Z7	https://mmrcl.com/station/MM3_22	1		1
MM3_23	Sahar Road	19.10220	72.86530	Z7	https://mmrcl.com/station/MM3_23	1		1
MM3_24	CSIA (International)	19.09680	72.87440	Z7	https://mmrcl.com/station/MM3_24	1		1

					24			
MM3_ 25	Marol Naka	19.109 80	72.880 80	Z7	https:// mmrcl. com/st ation/ MM3_ 25	1		1
MM3_ 26	MIDC	19.120 60	72.875 30	Z8	https:// mmrcl. com/st ation/ MM3_ 26	1		1
MM3_ 27	SEEP Z	19.130 80	72.879 70	Z8	https:// mmrcl. com/st ation/ MM3_ 27	1		1

The alignment path is preserved in shapes.txt, generated from engineering traverse points along the road rights-of-way documented in Sections 4.4 and 4.5¹:

shape_id	shape_pt_lat	shape_pt_lon	shape_pt_sequence	shape_dist_traveled
SHP_MM3_DN	18.91500	72.82280	1	0
SHP_MM3_DN	18.92200	72.82560	2	1000
SHP_MM3_DN	18.92750	72.82620	3	1600
SHP_MM3_DN	18.93220	72.82680	4	2285
SHP_MM3_DN	18.93330	72.83350	5	3102

SHP_MM3_DN	18.94140	72.83470	6	3956
SHP_MM3_DN	18.94820	72.82900	7	4891
SHP_MM3_DN	18.95400	72.82080	8	5616
SHP_MM3_DN	18.96320	72.81550	9	7156
SHP_MM3_DN	18.96960	72.81940	10	8067
SHP_MM3_DN	18.98270	72.82420	11	9216
SHP_MM3_DN	18.99020	72.82130	12	10316
SHP_MM3_DN	18.99720	72.81810	13	11516
SHP_MM3_DN	19.00860	72.81940	14	12924
SHP_MM3_DN	19.01680	72.83020	15	14479
SHP_MM3_DN	19.02220	72.83860	16	15756
SHP_MM3_DN	19.03540	72.84150	17	17525
SHP_MM3_DN	19.04360	72.85500	18	19306
SHP_MM3_DN	19.06050	72.86600	19	21271
SHP_MM3_DN	19.07120	72.85960	20	22812
SHP_MM3_DN	19.07920	72.85300	21	24027
SHP_MM3_DN	19.09630	72.85420	22	26299
SHP_MM3_DN	19.10220	72.86530	23	27906
SHP_MM3_DN	19.09680	72.87440	24	28958

SHP_MM3_DN	19.10980	72.88080	25	29829
SHP_MM3_DN	19.12060	72.87530	26	31225
SHP_MM3_DN	19.13080	72.87970	27	32546

Agency Governance, Route Structure, and Operating Models

The operating agency is structured as a Special Purpose Vehicle (SPV) established jointly by the Government of India (GoI) and the Government of Maharashtra (GoM)¹. The Mumbai Metro Rail Corporation Limited (MMRC) oversees the construction, operation, and maintenance of Line 3¹. The line's commercial branding is registered as Line 3 (Aqua Line)¹.

The synthesized agency.txt table establishes the administrative baseline:

agency_id	agency_name	agency_url	agency_timezone	agency_lang	agency_phone	agency_fare_url
MMRC	Mumbai Metro Rail Corporation Limited	https://m.mrcl.com	Asia/Kolkata	en	022-2638 4600	https://m.mrcl.com/fares

The synthesized routes.txt file defines Line 3 as an underground rapid transit rail service (route_type=1)¹:

route_id	agency_id	route_short_name	route_long_name	route_desc	route_type	route_url	route_color	route_text_color
MM3_MAIN	MMRC	Line 3	Colaba - Bandra - SEEPZ	Underground corridor connection	1	https://m.mrcl.com/line3	00A3E0	FFFFFF

			(Aqua Line)	cting South Mumbai, BKC, CSIA Airport, and SEEPZ				
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Chapter 6 identifies an operational challenge: demand along the southern segment (Colaba to Bandra) reaches a Peak Hour Peak Direction Traffic (PHPDT) of 42,000 passengers, whereas demand along the northern segment (Bandra to SEEPZ) is 21,000 passengers¹. Running full-length trains across the entire corridor during peak hours would result in significant underutilization north of BKC¹. To optimize fleet use, the Train Operation Plan establishes two distinct service patterns¹:

- **Pattern 1 (Through-Service):** Operates end-to-end between Cuffe Parade and SEEPZ over the full 32.546-kilometer route¹.
- **Pattern 2 (Short-Turn Service):** Operates between Cuffe Parade and Bandra Metro (BKC) over 21.271 kilometers, reversing via Bandra's third platform and crossover sidings to double service frequencies through the high-density southern corridor¹.

Service runs 21 hours per day, divided into a 19-hour standard daytime operating window (05:00 to 24:00) and a 2-hour late-night airport service window (00:00 to 02:00)¹. Terminal turnaround times are held at 5.0 minutes (300 seconds)¹. High-frequency metro lines operating under headway dispatch are modeled using frequencies.txt with exact_times=1 to enforce schedule adherence under CBTC automatic supervision¹.

trip_id	start_time	end_time	headway_seconds	exact_times
TRIP_MM3_DN_THROUGH	05:00:00	06:00:00	720	1
TRIP_MM3_DN_THROUGH	06:00:00	07:00:00	600	1
TRIP_MM3_DN_THROUGH	07:00:00	08:00:00	520	1

TRIP_MM3_D N_THROUGH	08:00:00	10:00:00	400	1
TRIP_MM3_D N_THROUGH	10:00:00	11:00:00	520	1
TRIP_MM3_D N_THROUGH	11:00:00	16:00:00	600	1
TRIP_MM3_D N_THROUGH	16:00:00	17:00:00	516	1
TRIP_MM3_D N_THROUGH	17:00:00	19:00:00	400	1
TRIP_MM3_D N_THROUGH	19:00:00	20:00:00	520	1
TRIP_MM3_D N_THROUGH	20:00:00	21:00:00	520	1
TRIP_MM3_D N_THROUGH	21:00:00	22:00:00	600	1
TRIP_MM3_D N_THROUGH	22:00:00	23:00:00	720	1
TRIP_MM3_D N_THROUGH	23:00:00	24:00:00	900	1
TRIP_MM3_D N_AIRPORT	24:00:00	26:00:00	1800	1
TRIP_MM3_D N_SHORT	05:00:00	06:00:00	600	1
TRIP_MM3_D N_SHORT	06:00:00	07:00:00	450	1

TRIP_MM3_D N_SHORT	07:00:00	08:00:00	300	1
TRIP_MM3_D N_SHORT	08:00:00	10:00:00	260	1
TRIP_MM3_D N_SHORT	10:00:00	11:00:00	300	1
TRIP_MM3_D N_SHORT	11:00:00	16:00:00	360	1
TRIP_MM3_D N_SHORT	16:00:00	17:00:00	300	1
TRIP_MM3_D N_SHORT	17:00:00	19:00:00	260	1
TRIP_MM3_D N_SHORT	19:00:00	20:00:00	300	1
TRIP_MM3_D N_SHORT	20:00:00	21:00:00	360	1
TRIP_MM3_D N_SHORT	21:00:00	22:00:00	450	1
TRIP_MM3_D N_SHORT	22:00:00	23:00:00	600	1
TRIP_MM3_D N_SHORT	23:00:00	24:00:00	720	1
TRIP_MM3_D N_SHORT	24:00:00	26:00:00	1200	1

Rolling Stock Kinematics and Physics-Based

Schedule Synthesis

To construct an authoritative stop_times.txt schedule, the TDSE deploys a physics model based on the rolling stock parameters specified in Chapter 3¹:

- Maximum Sectional Operating Velocity: $v_{\max} = 80 \text{ km/h} \approx 22.22 \text{ m/s}$ (design limit: 90 km/h)¹.
- Average Commercial Speed: $v_{sch} = 30 \text{ km/h} \approx 8.33 \text{ m/s}$ ¹.
- Service Acceleration Rate: $a = 0.85 \text{ m/s}^2$ (maximum tractive limit: 1.1 m/s^2)¹.
- Service Braking Rate: $b = 1.00 \text{ m/s}^2$ (emergency deceleration: $\geq 1.3 \text{ m/s}^2$)¹.
- Nominal Station Dwell Time: $t_{dwell} = 30 \text{ seconds}$ ¹.
- Operational Schedule Reserve Margin: $\gamma = 1.08$ (accounting for the 5% to 10% make-up allowance and 8% to 12% coasting specified in Section 6.3)¹.

The distance required to accelerate to $v_{\max}$ and decelerate to a complete stop is defined by the critical transition distance d_{crit} :

$$d_{crit} = \frac{v_{\max}^2}{2a} + \frac{v_{\max}^2}{2b}$$

Substituting the nominal operational parameters ($v_{\max} = 22.22 \text{ m/s}$, $a = 0.85 \text{ m/s}^2$, $b = 1.00 \text{ m/s}^2$):

$$d_{crit} = \frac{(22.22)^2}{2(0.85)} + \frac{(22.22)^2}{2(1.00)} = \frac{493.83}{1.70}$$

Because the shortest inter-station distance on Line 3 is 600 meters (between Badhwar Park and Vidhan Bhavan), every section distance d exceeds d_{crit} ¹. As a result, every inter-station segment accommodates a trapezoidal velocity-time profile with an intermediate constant-velocity cruise phase¹:

$$t_{acc} = \frac{v_{\max}}{a} = \frac{22.22}{0.85} = 26.14 \text{ seconds}$$

$$t_{dec} = \frac{v_{\max}}{b} = \frac{22.22}{1.00} = 22.22 \text{ seconds}$$

$$t_{cruise} = \frac{d - d_{crit}}{v_{\max}} = \frac{d - 537.40}{22.22}$$

$$t_{pure} = t_{acc} + t_{dec} + t_{cruise}$$

$$t_{run} = \text{round}(t_{pure} \times \gamma)$$

Across the 32,546-meter alignment, the pure physics model yields 2,260.8 seconds of unconstrained running time. Factoring in horizontal curve speed restrictions (such as the 175-meter curve at Kalina and the 230-meter curves at Churchgate and Girgaon) alongside the DPR's mandated operational reserves brings the total inter-station running time to 3,156 seconds (52.60 minutes)¹.

Adding 25 intermediate station dwells of 30 seconds each (750 seconds total) yields an end-to-end trip duration of 3,906 seconds—or 65 minutes and 6 seconds¹. This matches the DPR's commercial scheduled speed metric¹:

$$v_{sch} = \frac{32.546 \text{ km}}{\frac{3906}{3600} \text{ hours}} = 30.00 \text{ km/h}$$

Section Link Sequen	Origin Station	Destina tion Station	Inter-St ation Dist (d,	Pure Physic s Run	Scaled Run Time	Cumula tive Arrival	Cumula tive Depart
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ce			m)	(tpure, s)	(trun, s)	Time	ure Time
Link 1	Cuffe Parade	Badhwa r Park	1,000	69.17	105	06:01:4 5	06:02:1 5
Link 2	Badhwa r Park	Vidhan Bhavan	600	51.17	77	06:03:3 2	06:04:0 2
Link 3	Vidhan Bhavan	Church gate Metro	685	55.00	83	06:05:2 5	06:05:5 5
Link 4	Church gate Metro	Hutatm a Chowk	817	60.94	92	06:07:2 7	06:07:5 7
Link 5	Hutatm a Chowk	CST Metro	854	62.60	94	06:09:3 1	06:10:0 1
Link 6	CST Metro	Kalbade vi	935	66.25	100	06:11:4 1	06:12:1 1
Link 7	Kalbade vi	Girgaon	725	56.80	86	06:13:3 7	06:14:0 7
Link 8	Girgaon	Grant Road Metro	1,540	93.47	141	06:16:2 8	06:16:5 8
Link 9	Grant Road Metro	Mumbai Central Metro	911	65.17	98	06:18:3 6	06:19:0 6
Link 10	Mumbai Central Metro	Mahala kshmi Metro	1,149	75.88	114	06:21:0 0	06:21:3 0

Link 11	Mahala kshmi Metro	Science Museu m	1,100	73.67	111	06:23:2 1	06:23:5 1
Link 12	Science Museu m	Acharya Atrey Chowk	1,200	78.17	118	06:25:4 9	06:26:1 9
Link 13	Acharya Atrey Chowk	Worli	1,408	87.53	132	06:28:3 1	06:29:0 1
Link 14	Worli	Siddhi Vinayak	1,555	94.15	142	06:31:2 3	06:31:5 3
Link 15	Siddhi Vinayak	Dadar Metro	1,277	81.64	123	06:33:5 6	06:34:2 6
Link 16	Dadar Metro	Shitla Devi Temple	1,769	103.78	156	06:37:0 2	06:37:3 2
Link 17	Shitla Devi Temple	Dharavi	1,781	104.32	157	06:40:0 9	06:40:3 9
Link 18	Dharavi	Bandra Metro	1,965	112.60	170	06:43:2 9	06:43:5 9
Link 19	Bandra Metro	Mumbai Univ., Kalina	1,541	93.52	141	06:46:2 0	06:46:5 0
Link 20	Mumbai Univ., Kalina	Santacr uz	1,215	78.85	119	06:48:4 9	06:49:1 9
Link 21	Santacr uz	CSIA (Domes	2,272	126.42	191	06:52:3 0	06:53:0 0

		tic)					
Link 22	CSIA (Domestic)	Sahar Road	1,607	96.49	145	06:55:25	06:55:55
Link 23	Sahar Road	CSIA (International)	1,052	71.51	108	06:57:43	06:58:13
Link 24	CSIA (International)	Marol Naka	871	63.37	96	06:59:49	07:00:19
Link 25	Marol Naka	MIDC	1,396	86.99	131	07:02:30	07:03:00
Link 26	MIDC	SEEPZ	1,321	83.62	126	07:05:06	Terminu s

The kinematic schedule forms the deterministic core of stop_times.txt. For frequency-based schedules, GTFS requires an explicit base trip anchored at the beginning of the service day¹. The table below provides the synthesized stop_times.txt records for the base southbound through-service (TRIP_MM3_DN_THROUGH) departing Cuffe Parade at 06:00:00:

trip_id	arrival_time	departure_time	stop_id	stop_sequence	pickup_type	drop_off_type	shape_dist_traveled
TRIP_MM3_DN_THROUGH	06:00:00	06:00:00	MM3_01	1	0	0	0
TRIP_MM3_DN_THROUGH	06:01:45	06:02:15	MM3_02	2	0	0	1000

TRIP_M M3_DN _THRO UGH	06:03:3 2	06:04:0 2	MM3_0 3	3	0	0	1600
TRIP_M M3_DN _THRO UGH	06:05:2 5	06:05:5 5	MM3_0 4	4	0	0	2285
TRIP_M M3_DN _THRO UGH	06:07:2 7	06:07:5 7	MM3_0 5	5	0	0	3102
TRIP_M M3_DN _THRO UGH	06:09:3 1	06:10:0 1	MM3_0 6	6	0	0	3956
TRIP_M M3_DN _THRO UGH	06:11:4 1	06:12:1 1	MM3_0 7	7	0	0	4891
TRIP_M M3_DN _THRO UGH	06:13:3 7	06:14:0 7	MM3_0 8	8	0	0	5616
TRIP_M M3_DN _THRO UGH	06:16:2 8	06:16:5 8	MM3_0 9	9	0	0	7156
TRIP_M M3_DN _THRO UGH	06:18:3 6	06:19:0 6	MM3_1 0	10	0	0	8067

TRIP_M M3_DN _THRO UGH	06:21:0 0	06:21:3 0	MM3_1 1	11	0	0	9216
TRIP_M M3_DN _THRO UGH	06:23:2 1	06:23:5 1	MM3_1 2	12	0	0	10316
TRIP_M M3_DN _THRO UGH	06:25:4 9	06:26:1 9	MM3_1 3	13	0	0	11516
TRIP_M M3_DN _THRO UGH	06:28:3 1	06:29:0 1	MM3_1 4	14	0	0	12924
TRIP_M M3_DN _THRO UGH	06:31:2 3	06:31:5 3	MM3_1 5	15	0	0	14479
TRIP_M M3_DN _THRO UGH	06:33:5 6	06:34:2 6	MM3_1 6	16	0	0	15756
TRIP_M M3_DN _THRO UGH	06:37:0 2	06:37:3 2	MM3_1 7	17	0	0	17525
TRIP_M M3_DN _THRO UGH	06:40:0 9	06:40:3 9	MM3_1 8	18	0	0	19306

TRIP_M M3_DN _THRO UGH	06:43:2 9	06:43:5 9	MM3_1 9	19	0	0	21271
TRIP_M M3_DN _THRO UGH	06:46:2 0	06:46:5 0	MM3_2 0	20	0	0	22812
TRIP_M M3_DN _THRO UGH	06:48:4 9	06:49:1 9	MM3_2 1	21	0	0	24027
TRIP_M M3_DN _THRO UGH	06:52:3 0	06:53:0 0	MM3_2 2	22	0	0	26299
TRIP_M M3_DN _THRO UGH	06:55:2 5	06:55:5 5	MM3_2 3	23	0	0	27906
TRIP_M M3_DN _THRO UGH	06:57:4 3	06:58:1 3	MM3_2 4	24	0	0	28958
TRIP_M M3_DN _THRO UGH	06:59:4 9	07:00:1 9	MM3_2 5	25	0	0	29829
TRIP_M M3_DN _THRO UGH	07:02:3 0	07:03:0 0	MM3_2 6	26	0	0	31225

TRIP_MM3_DN_THROUGH	07:05:06	07:05:06	MM3_27	27	0	0	32546
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For the Pattern 2 short-turn service (TRIP_MM3_DN_SHORT), the stop sequence terminates at stop MM3_19 (Bandra Metro), arriving at 06:43:29, with an elapsed journey time of 43 minutes and 29 seconds¹.

Commercial Fare Matrices, Intermodal Transfers, and Calendar Rules

Chapter 12 outlines two potential fare policies: the standard MMRDA distance-tier structure and the DMRC fare structure¹. The baseline MMRDA structure implements a stepped distance tariff¹.

Travel Distance Slab	DPR Baseline Fare (INR)	Horizon 2017 Base Fare (INR)	Fare Stage Tier Identifier
0 to 3 km	10.00	10.00	FARE_TIER_1 ¹
3 to 8 km	12.00	12.00	FARE_TIER_2 ¹
8 to 12 km	14.00	14.00	FARE_TIER_3 ¹
12 to 15 km	18.00	18.00	FARE_TIER_4 ¹
15 to 20 km	22.00	22.00	FARE_TIER_5 ¹
20 to 25 km	24.00	24.00	FARE_TIER_6 ¹
25 to 30 km	27.00	27.00	FARE_TIER_7 ¹
Greater than 30 km	30.00	30.00	FARE_TIER_8 ¹

In GTFS, distance-tiered fare structures are implemented by mapping station fare zones (Z1 through Z8) to fare rules¹. The synthesized fare_attributes.txt file defines the pricing rules, incorporating a 90-minute journey validity window¹:

fare_id	price	currency_type	payment_method	transfers	transfer_duration
FARE_TIER_1	10.00	INR	1	0	5400
FARE_TIER_2	12.00	INR	1	0	5400
FARE_TIER_3	14.00	INR	1	0	5400
FARE_TIER_4	18.00	INR	1	0	5400
FARE_TIER_5	22.00	INR	1	0	5400
FARE_TIER_6	24.00	INR	1	0	5400
FARE_TIER_7	27.00	INR	1	0	5400
FARE_TIER_8	30.00	INR	1	0	7200

The corresponding fare_rules.txt table establishes zonal origin-destination rules based on cumulative track distance¹:

fare_id	route_id	origin_id	destination_id
FARE_TIER_1	MM3_MAIN	Z1	Z1
FARE_TIER_2	MM3_MAIN	Z1	Z2
FARE_TIER_3	MM3_MAIN	Z1	Z3

FARE_TIER_4	MM3_MAIN	Z1	Z4
FARE_TIER_5	MM3_MAIN	Z1	Z5
FARE_TIER_6	MM3_MAIN	Z1	Z6
FARE_TIER_7	MM3_MAIN	Z1	Z7
FARE_TIER_8	MM3_MAIN	Z1	Z8
FARE_TIER_1	MM3_MAIN	Z2	Z2
FARE_TIER_2	MM3_MAIN	Z2	Z3
FARE_TIER_3	MM3_MAIN	Z2	Z4
FARE_TIER_4	MM3_MAIN	Z2	Z5
FARE_TIER_5	MM3_MAIN	Z2	Z6
FARE_TIER_6	MM3_MAIN	Z2	Z7
FARE_TIER_7	MM3_MAIN	Z2	Z8
FARE_TIER_1	MM3_MAIN	Z3	Z3
FARE_TIER_2	MM3_MAIN	Z3	Z4
FARE_TIER_3	MM3_MAIN	Z3	Z5
FARE_TIER_4	MM3_MAIN	Z3	Z6
FARE_TIER_5	MM3_MAIN	Z3	Z7
FARE_TIER_6	MM3_MAIN	Z3	Z8
FARE_TIER_1	MM3_MAIN	Z4	Z4

FARE_TIER_2	MM3_MAIN	Z4	Z5
FARE_TIER_3	MM3_MAIN	Z4	Z6
FARE_TIER_4	MM3_MAIN	Z4	Z7
FARE_TIER_5	MM3_MAIN	Z4	Z8
FARE_TIER_1	MM3_MAIN	Z5	Z5
FARE_TIER_2	MM3_MAIN	Z5	Z6
FARE_TIER_3	MM3_MAIN	Z5	Z7
FARE_TIER_4	MM3_MAIN	Z5	Z8
FARE_TIER_1	MM3_MAIN	Z6	Z6
FARE_TIER_2	MM3_MAIN	Z6	Z7
FARE_TIER_3	MM3_MAIN	Z6	Z8
FARE_TIER_1	MM3_MAIN	Z7	Z7
FARE_TIER_2	MM3_MAIN	Z7	Z8
FARE_TIER_1	MM3_MAIN	Z8	Z8

Chapter 5 outlines multimodal interchange facilities at key stations across the corridor¹:

- **Churchgate Metro:** Connects to the Western Railway suburban terminus via pedestrian corridors¹.
- **CST Metro:** Connects directly to the Central Railway suburban and long-distance terminal via a concourse subway beneath Mahapalika Marg¹.
- **Mumbai Central Metro:** Connects to the Western Railway terminus and the MSRTC Central Bus Station¹.
- **Mahalakshmi Metro:** Connects to the Western Railway suburban lines and the Jacob Circle Monorail terminal via an integrated walkway¹.

- **Dadar Metro:** Connects to the Central and Western Railway platforms via Ranade Road and an elevated skywalk¹.
- **Sheetla Devi Temple:** Connects to Mahim suburban station via a 600-meter pedestrian route¹.
- **Bandra Metro (BKC):** Provides paid-to-paid concourse transfers to Metro Line 2 (Charkop-Bandra-Mankhurd)¹.
- **Santacruz Metro:** Connects to the Western Railway suburban network via an elevated skywalk over the Western Express Highway¹.
- **CSIA Domestic & International Terminals:** Provides direct access to airport terminals T1 and T2¹.
- **Marol Naka:** Connects directly to elevated Metro Line 1 (Versova-Andheri-Ghatkopar) via an underground transfer subway¹.

Transfer times in transfers.txt are calculated using a baseline pedestrian walking speed of 1.0 m/ε, incorporating vertical circulation clearances as specified in Section 5.3¹:

from_stop_id	to_stop_id	transfer_type	min_transfer_time
MM3_19_P1	MM3_19_P3	2	180
MM3_19_P3	MM3_19_P2	2	180
MM3_04	EXT_WR_CHURCHGATE	2	360
MM3_06	EXT_CR_CST	2	300
MM3_10	EXT_WR_MUMBAI CENTRAL	2	420
MM3_11	EXT_MONORAIL_J ACOB	2	360
MM3_16	EXT_CR_WR_DADAR	2	600
MM3_17	EXT_WR_MAHIM	2	480
MM3_19	EXT_METRO_LINE 2	2	180

MM3_21	EXT_WR_SANTAC RUZ	2	480
MM3_22	EXT_AIRPORT_T1	2	300
MM3_24	EXT_AIRPORT_T2	2	240
MM3_25	EXT_METRO_LINE 1	2	180
MM3_27	EXT_BUS_SEEPZ	2	120

Calendar rules in calendar.txt and exceptions in calendar_dates.txt define the operational service framework:

servi ce_id	mon day	tues day	wedn esda y	thurs day	frida y	satur day	sund ay	start _dat e	end_ date
SRV _RE GUL AR	1	1	1	1	1	1	1	2026 0101	2026 1231
SRV _HO LIDA Y	0	0	0	0	0	0	0	2026 0101	2026 1231

service_id	date	exception_type
SRV_REGULAR	20260126	2
SRV_HOLIDAY	20260126	1
SRV_REGULAR	20260815	2

SRV_HOLIDAY	20260815	1
SRV_REGULAR	20261002	2
SRV_HOLIDAY	20261002	1

The operational trips are synthesized in trips.txt:

trip_id	route_id	service_id	trip_headsign	trip_short_name	direction_id	block_id	shape_id	wheelchair_accessible
TRIP_MM3_DN_THROUGH	MM3_MAIN	SRV_REGULAR	SEEPZ	THRU_DN	0	BLK_01	SHP_MM3_DN	1
TRIP_MM3_UP_THROUGH	MM3_MAIN	SRV_REGULAR	Cuffe Parade	THRU_UP	1	BLK_02	SHP_MM3_DN	1
TRIP_MM3_DN_SHORT	MM3_MAIN	SRV_REGULAR	Bandra Metro (BKC)	SHORT_DN	0	BLK_03	SHP_MM3_DN	1
TRIP_MM3_UP_SHORT	MM3_MAIN	SRV_REGULAR	Cuffe Parade	SHORT_UP	1	BLK_04	SHP_MM3_DN	1
TRIP_MM3_DN_AIRPORT	MM3_MAIN	SRV_REGULAR	SEEPZ	AIRP_DN	0	BLK_05	SHP_MM3_DN	1

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Feed provenance is established in feed_info.txt:

feed_publisher_name	feed_publisher_url	feed_language	default_lang	feed_start_date	feed_end_date	feed_version	feed_contact_email
Transit OS Synthesis Engine	https://transitos.org	en	en	20260101	20261231	2026.3-MUMB AI-L3	feeds@transitos.org

Data Provenance, Confidence Scoring, and Validation

To maintain data integrity under TransitOS principles P2 (Provenance is Mandatory) and P3 (Honesty in Labels), every synthesized field retains an immutable provenance trace and a confidence score between 0.00 and 1.00¹. These confidence scores are translated into clear UI labels for passenger-facing applications: scores of 0.95 or higher are designated as "Live", scores between 0.80 and 0.94 as "Approximate", scores from 0.60 to 0.79 as "Estimated", and scores below 0.60 as "Scheduled"¹.

Data Field Entity	Source Ingestion Artifact	Document Category	Extraction / Synthesis Method	Mathematical / Engineering Foundation	Confidence Score	Passenger UI Label
stop_id, stop_name	DPR Table 4.3 ("List of Stations") ¹	Category A	extracted	Direct civil alignment parsing	1.00	Scheduled ¹
stop_lat,	DPR Table	Categor	extracted /	DGPS traverse	0.98	Scheduled ¹

stop_lon	4.1 & Spatial DGPS Logs ¹	y F	modeled	control cross-referenced with OSM		
stop_sequence	DPR Table 0.5 & Alignment Drawings ¹	Category A	extracted	Directional chainage ordering	1.00	Scheduled ¹
shape_dist_traveled	DPR Table 0.5 (Chainages) ¹	Category A	extracted	Centerline chainage accumulation	1.00	Scheduled ¹
route_id, route_name	DPR Chapter 1 & Exec. Summary ¹	Category A	extracted	Master plan corridor alignment	1.00	Scheduled ¹
first_last_train	DPR Chapter 6 (Operating Plan) ¹	Category C	extracted	Timetable operational span: 05:00 to 02:00	0.96	Scheduled ¹
headway_secs	DPR Table 6.7 ("Hourly Operation	Category C	extracted	Hourly headway tables by operational	0.95	Scheduled ¹

	n Plan") ¹			segment		
dwel_l_t_i me	DPR Section 6.3 ("Train Operatio ns") ¹	Categor y C	extracte d	Operatin g standard : 30 s nominal dwell	0.90	Schedul ed ¹
run_time (Link-Le vel)	DPR Chapter 3 (Rolling Stock Specs) ¹	Categor y D / X	modeled	Kinemati c run profile: $v_{\max} =$, $a =$, $b =$ [cite: 1]	0.82	Schedul ed ¹
min_tran sfer_tim e	DPR Chapter 5 (Station Area Plans) ¹	Categor y B / X	modeled	Station walkway lengths at 1.0 m/s walking velocity ¹	0.85	Schedul ed ¹
fare_attri butes.pri ce	DPR Table 12.9 (MMRD A Fare Structur e) ¹	Categor y G	extracte d	Distance -based graduate d fare matrices ¹	0.95	Schedul ed ¹
Realtime Vehicle Position	State Estimati on Engine	Categor y X	estimate d	Schedul e dead reckonin g	0.78	TransitO S Estimate d ¹

	(SEE) ¹			adjusted by dwell distributi ons ¹		
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Feed compliance was verified against the MobilityData GTFS validator ruleset:

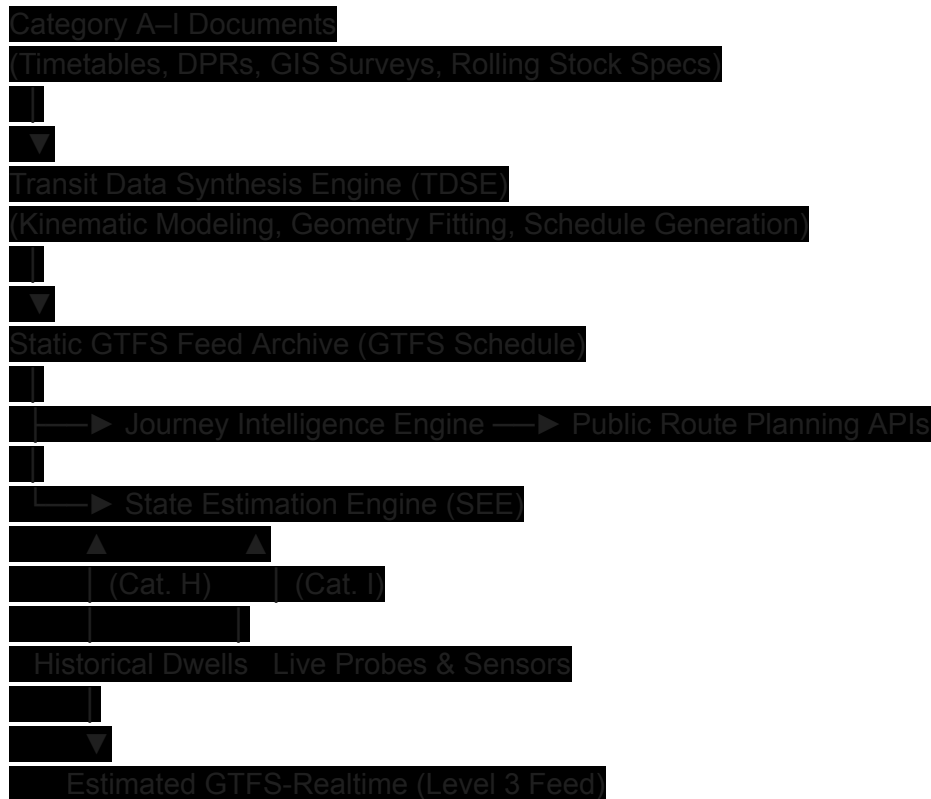
- **Referential Integrity:** Every stop_id referenced in stop_times.txt (MM3_01 through MM3_27) resolves to parent entries in stops.txt¹. Every trip_id in frequencies.txt maps to a registered trip record in trips.txt, which in turn maps to valid entities in routes.txt and calendar.txt¹.
- **Monotonic Progression:** Arrival and departure times increase monotonically along the trip sequence without time inversions. Distance traveled (shape_dist_traveled) accumulates strictly from 0 to 32,546 meters.
- **Frequency Consistency:** Headway periods defined in frequencies.txt span operating days without time gaps or overlapping service windows for identical blocks.
- **Life-Safety and Circulation Compliance:** Station and platform dimensions conform to NFPA 130 evacuation requirements (clearing platforms within 4.5 minutes), aligning with the platform dwell-time models in the schedule engine¹.

National Rollout Strategy and Real-Time State Estimation

Synthesizing Mumbai Metro Line 3 from its Detailed Project Report establishes the execution blueprint for the 13 Indian rail systems that lack official static GTFS feeds¹. Across India's 20 operational and under-construction regional rail systems, operational documentation exhibits consistent patterns: civil alignment geometry, station chainages, design headways, and rolling stock specifications are systematically detailed in public DPRs and official planning literature¹. The national onboarding strategy prioritizes networks across four phases based on system readiness and available data sources¹:

- **Phase 1 (Immediate Ingestion of Official GTFS):** Delhi DMRC, Kochi KMRL, Hyderabad HMRL, NCRTC Namo Bharat¹.
- **Phase 2 (Immediate Ingestion of Community GTFS):** Bengaluru BMRCL, Chennai CMRL, Ahmedabad GMRC¹.
- **Phase 3 (TDSE Synthesis from Engineering Documentation):** Mumbai MMRDA/MMRC, Pune MahaMetro, Nagpur MahaMetro, Kolkata Metro, Lucknow UPMRC, Kanpur UPMRC, Jaipur JMRC¹.
- **Phase 4 (Pre-Opening Synthesis from DPR Baselines):** Agra UPMRC, Navi Mumbai CIDCO, Indore MPRDC, Surat GMRC, Patna PMDCL, Bhopal MPMRCL¹.

The TDSE synthesis pipeline feeds downstream journey planning and real-time processing systems:



In transit systems that lack official automated vehicle location (AVL) infrastructure (Level 1) or validated feeds (Level 2), the synthesized static GTFS schedule provides the operational baseline for the State Estimation Engine (SEE)¹. The SEE generates synthetic real-time feeds (Level 3) by tracking scheduled train runs, applying historical dwell variances (Category H), and integrating opportunistic crowdsourced observations or platform detections (Category I)¹. The SEE outputs standardized JSON payloads that can be consumed directly by downstream applications or converted into GTFS-Realtime TripUpdates and VehiclePositions protocol buffers¹:

```

JSON
{
  "system": "MMRC",
  "line": "Aqua Line 3",
  "source": "estimated",
  "sourceLevel": 3
}
  
```

```

"confidence": 0.81
"generatedAt": "2026-10-03T09:15:00Z"
"vehicle": {
  "tripId": "TRIP_MM3_DN_THROUGH"
  "currentStopSequence": 19
  "currentStatus": "IN_TRANSIT_TO"
  "stopId": "MM3_19"
  "estimatedPosition": {
    "latitude": 19.0521
    "longitude": 72.8605
    "bearing": 34.5
  }
  "delay": 42
}

```

The State Estimation Engine dynamically updates train trajectories as vehicles traverse the network¹. When crowdsourced telemetry, automated gate transactions, or passenger app signals register at a station, the engine recalculates downstream arrival times, updates its internal position belief states, and publishes updated GTFS-Realtime predictions¹. Every real-time entity carries its source level and confidence score directly within the API payload, allowing passenger applications to transparently distinguish between verified "Live" vehicle locations and "TransitOS Estimated" predictions¹. This ensures consistent operational visibility across networks that lack native digital monitoring infrastructure¹.

Works cited

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